

TASK 4 – REPORT

Table of valid observations:

Location	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Healthy Forest	21	32	44	40	30	60	49	64	46	36	16	28
Dense Forest 1	9	19	22	22	16	23	29	25	32	18	7	7
Dense Forest 2	3	11	22	24	18	28	27	29	30	22	6	3
Medium Forest	5	15	20	27	19	26	28	27	27	15	6	5
Sparse Forest	8	23	21	24	20	25	28	26	28	19	7	6
Very Sparse Forest	8	20	21	23	18	24	28	33	33	19	11	7

Simple Heatmap (Green-Yellow-Red Colour Scale):

Location	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Healthy Forest	21	32	44	40	30	60	49	64	46	36	16	28
Dense Forest 1	9	19	22	22	16	23	29	25	32	18	7	7
Dense Forest 2	3	11	22	24	18	28	27	29	30	22	6	3
Medium Forest	5	15	20	27	19	26	28	27	27	15	6	5
Sparse Forest	8	23	21	24	20	25	28	26	28	19	7	6
Very Sparse Forest	8	20	21	23	18	24	28	33	33	19	11	7

Observation:

The months with the highest observation counts (approximately June-September) correspond to the main growing season, where vegetation is most active. Consequently, the NDVI values and disturbance signals observed during these months are based on a relatively large number of cloud-free observations, increasing confidence in the conclusions drawn in Tasks 1-3.

Conversely, November through January contain substantially fewer valid observations due to frequent cloud cover, snow, and reduced daylight during winter.

High Confidence: As noted, the months spanning June through September show the highest volume of valid satellite observations across all pixels (peaking at 60 and 64 observations for the Healthy Forest baseline). This corresponds perfectly with the peak growing season. The sudden drop observed during the May-July 2022 clearing event in Dense Forest 1 can be accepted with maximum confidence.

Low Confidence: Conversely, the window from November through January has sparse data. Valid observations are observed as single digits. This data sparsity perfectly explains the strange behaviour evaluated in Sparse Forest (Pixel 4), where the residual plummeted to -0.473 in December 2022 before violently bouncing back to a highly positive +0.276 in January 2023.

*The data audit confirms that our summer deforestation tracking is highly dependable. However, because winter data is so limited, any sudden drops or strange anomalies appearing between November and January should be treated with caution.

GEE Code Link: <https://code.earthengine.google.com/9fda3427466107ecb8406eec3c1f4d27>

GitHub Repository Link: <https://github.com/SiddharthBhimpure/bavaria-forest-change-analysis.git>

END OF REPORT